PHARMACOLOGY · MCQ PRACTICE

Drugs Used in Infertility & Assisted Reproduction — MCQ Practice

SGL 2 · Dr. Chro Yaseen Qader
GUE BLOCK

Pharmacology GUE Block · MEDucation by Mattin Majid

HOW TO USE THIS SET
20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.
4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. By the lecture's definition, infertility is diagnosed after what period of regular unprotected intercourse without conception? EASY

A. 6 months
B. 3 months
C. 2 years
D. 1 year
E. 5 years

2. Which of the following is a recognised female cause of infertility? EASY

A. Cryptorchidism
B. Varicocele
C. Oligospermia
D. Azoospermia
E. Endometriosis

3. Clomiphene citrate's mechanism of action is best described as: EASY

A. An aromatase inhibitor
B. A competitive antagonist at hypothalamic/pituitary oestrogen receptors
C. A dopamine receptor agonist
D. A direct agonist at ovarian FSH receptors
E. A progesterone receptor partial agonist

4. Which drug class is a non-steroidal reversible competitive inhibitor of the enzyme that converts androstenedione and testosterone to oestrogens? EASY

A. SERMs
B. Aromatase inhibitors
C. Gonadotropins
D. GnRH agonists
E. Dopamine agonists

5. A patient is being tested with the Clomiphene Citrate Challenge Test. What is this test primarily used to assess? MEDIUM

A. Male sperm motility
B. Ovarian reserve
C. Adrenal reserve
D. Thyroid function
E. Pituitary prolactin secretion

6. A 32-year-old with PCOS fails to ovulate on clomiphene. Which drug class would reasonably be tried next for anovulatory infertility? MEDIUM

- **A.** Aromatase inhibitors only, never gonadotropins
- **B.** Antiprogestogens
- **C.** Dopamine agonists
- **D.** Tamoxifen
- **E.** Gonadotropins (hMG/FSH)

7. In an IVF cycle, why is a GnRH agonist typically given continuously BEFORE gonadotropin stimulation rather than during it? MEDIUM

- **A.** Because continuous GnRH triggers ovulation
- **B.** To reduce the dose of gonadotropins needed
- **C.** To directly increase follicle number
- **D.** To suppress endogenous FSH/LH so a premature LH surge doesn't trigger early ovulation
- **E.** To replace the need for hCG entirely

8. Urofollitropin (uFSH) differs from recombinant follitropin alfa/beta mainly in: MEDIUM

- **A.** Being effective only in males
- **B.** Being a dopamine agonist rather than a gonadotropin
- **C.** Its source — extracted from the urine of postmenopausal women rather than produced recombinantly
- **D.** Requiring oral administration
- **E.** Acting on a completely different receptor

9. A woman undergoing ovulation induction is given hCG after a course of gonadotropins. What is the purpose of the hCG? MEDIUM

- **A.** To mimic the LH surge and trigger ovulation
- **B.** To treat ovarian hyperstimulation syndrome
- **C.** To suppress FSH secretion
- **D.** To block the oestrogen receptor
- **E.** To inhibit prolactin secretion

10. Which hormone receptor is shared by LH and hCG, allowing the two to be used interchangeably? MEDIUM

- **A.** The LH receptor
- **B.** The dopamine D2 receptor
- **C.** The oestrogen receptor
- **D.** The FSH receptor
- **E.** The GnRH receptor

11. A man with hypogonadotrophic hypogonadism and cryptorchidism is treated with gonadotropins. What is the therapeutic goal? MEDIUM

- **A.** To suppress FSH receptor activity
- **B.** To reduce testosterone levels
- **C.** To treat erectile dysfunction directly
- **D.** To correct infertility from a low sperm count due to inadequate gonadotropin drive
- **E.** To block spermatogenesis

12. Continuous GnRH-agonist therapy (e.g. nasal spray or depot/implant) is used for all of the following conditions EXCEPT: HARD

A. Breast cancer
B. Prostate cancer
C. Endometriosis
D. Hypothalamic hypogonadotropic hypogonadism requiring ovulation induction
E. Precocious puberty

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13. A patient with hyperprolactinemia and galactorrhea is being considered for bromocriptine or cabergoline. Which statement correctly compares the two? MEDIUM

A. Both are equally ineffective for this indication
B. Bromocriptine is a GnRH antagonist
C. Bromocriptine is more effective with fewer adverse effects
D. Cabergoline is more effective at resolving amenorrhea/galactorrhea with fewer GI adverse effects
E. Cabergoline works by blocking aromatase, unlike bromocriptine

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14. What is the mechanism by which bromocriptine and cabergoline lower prolactin levels? MEDIUM

A. Inhibition of TRH synthesis
B. Direct destruction of lactotroph cells
C. Blockade of the prolactin receptor
D. Oestrogen receptor blockade
E. D2 dopamine receptor agonism, inhibiting prolactin production

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15. Which adverse effect is specifically associated with gonadotropin therapy for ovulation induction, distinguishing it from clomiphene therapy's typical risks? MEDIUM

A. Ovarian hyperstimulation syndrome with ascites and hydrothorax
B. Peripheral neuropathy
C. Hepatotoxicity
D. QT prolongation
E. Hot flushes alone

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16. Before starting continuous GnRH-agonist therapy in a man with bone-metastatic prostate cancer, why is an androgen-receptor antagonist such as flutamide given first? HARD

A. Because flutamide blocks the LH receptor directly
B. Because flutamide prevents ovarian hyperstimulation
C. Because GnRH agonists are ineffective without flutamide
D. Because flutamide is required to activate the GnRH agonist
E. Because the initial flare in gonadotropin/testosterone secretion can transiently worsen bone-metastasis pain

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17. FSH, LH and hCG are structurally related glycoproteins. What underlies their shared and distinct receptor actions? HARD

A. They are identical molecules with different names
B. They share an identical β chain but differ in the α chain
C. They all act exclusively through the FSH receptor
D. FSH and LH are proteins while hCG is a steroid
E. They share an identical α chain but have a distinct β chain that confers receptor specificity

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18. A patient on gonadotropin therapy for IVF develops severe abdominal pain, and imaging reveals free fluid in the peritoneal cavity after a large ovarian cyst was noted on prior scan. Which complication of gonadotropin therapy does this most likely represent? HARD

A. Arterial thromboembolism
B. Haemoperitoneum from a ruptured ovarian cyst
C. Hyperprolactinemia
D. GnRH-agonist flare
E. Aromatase inhibitor toxicity

19. Why is a GnRH agonist alone NOT the drug used to directly stimulate ovulation during a standard IVF cycle, even though pulsatile GnRH can induce ovulation in hypothalamic hypogonadism? HARD

A. In IVF protocols, GnRH agonists/antagonists are used continuously to SUPPRESS the LH surge, while ovulation is induced separately with gonadotropins and triggered with hCG
B. GnRH agonists cannot be administered by injection
C. GnRH agonists destroy ovarian follicles
D. GnRH agonists work only in men
E. GnRH agonists directly cause ovarian hyperstimulation syndrome and are avoided entirely in IVF

20. Multiple pregnancy risk with gonadotropin-induced ovulation is estimated at roughly what rate, compared with the general population? HARD

A. 15–20%, versus about 1% in the general population
B. Less than the general population
C. Only relevant for IVF, never for ovulation induction alone
D. 40–50%, versus about 10% in the general population
E. Equal to the general population at ~1%

A Answer key

#	Ans	Diff	Rationale
1	**D**	EASY	The lecture defines infertility as failure to conceive after 1 year of regular unprotected sex.
2	**E**	EASY	Endometriosis is listed as a female cause; azoospermia, cryptorchidism, varicocele and oligospermia are male causes.
3	**B**	EASY	Clomiphene is a competitive antagonist blocking hypothalamic/pituitary oestrogen receptors, removing negative feedback.
4	**B**	EASY	This describes aromatase inhibitors (anastrozole, letrozole), which block the androgen→oestrogen conversion step.
5	**B**	MEDIUM	The Clomiphene Citrate Challenge Test is a diagnostic use of clomiphene, assessing ovarian reserve.
6	**E**	MEDIUM	Gonadotropins are used for infertility caused by lack of ovulation following clomiphene failure.
7	**D**	MEDIUM	Continuous GnRH agonist suppresses endogenous FSH/LH before gonadotropins are given, preventing a premature LH surge.
8	**C**	MEDIUM	uFSH is extracted from postmenopausal urine, distinguishing it from the recombinant rFSH forms (follitropin alfa/beta).
9	**A**	MEDIUM	hCG mimics the LH surge via the shared LH receptor, triggering ovulation after follicular development.
10	**A**	MEDIUM	LH and hCG are mediated by the same LH receptor, making them interchangeable in clinical use.
11	**D**	MEDIUM	Gonadotropins address low sperm count from hypogonadotrophic hypogonadism/cryptorchidism by restoring gonadotropin drive.
12	**D**	HARD	Pulsatile (not continuous) GnRH is what's used to induce ovulation/spermatogenesis in hypothalamic hypogonadotropic hypogonadism; continuous GnRH is used for the other four (suppressive) indications.
13	**D**	MEDIUM	Cabergoline resolves amenorrhea/oligomenorrhea due to galactorrhea more effectively than bromocriptine, with significantly fewer GI adverse events.
14	**E**	MEDIUM	Both bromocriptine and cabergoline are D2 receptor agonists that inhibit prolactin production.

15	**A**	MEDIUM	Ovarian hyperstimulation syndrome (ovarian enlargement, ascites, hydrothorax, hypovolemia) is the characteristic serious complication of gonadotropin therapy.
16	**E**	HARD	The initial gonadotropin/testosterone flare on starting a GnRH agonist can transiently worsen bone-metastasis pain in prostate cancer, so an androgen-receptor antagonist is given first to blunt it.
17	**E**	HARD	FSH, LH and hCG share an identical α chain; a distinct β chain confers each hormone's receptor specificity (FSH receptor vs shared LH receptor for LH/hCG).
18	**B**	HARD	Abdominal pain with free peritoneal fluid after a known ovarian cyst points to haemoperitoneum from a ruptured ovarian cyst — one of the gonadotropin therapy complications listed.
19	**A**	HARD	In IVF, GnRH agonists/antagonists are used continuously to suppress the LH surge; gonadotropins stimulate follicular growth, and hCG/rhCG is given separately to trigger ovulation.
20	**A**	HARD	The lecture states the risk of multiple pregnancy with gonadotropin therapy is 15–20%, compared with roughly 1% in the general population.